Mathematical Review Zyklus23

Review date: 4 November 2026 · English translation
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Mathematical development, implementation and initial verification of the original project: Claude (Anthropic)
Additional mathematical review and development proposals: ChatGPT (OpenAI)

Assessment

The exact calendar arithmetic is sound. Rounding errors remain bounded relative to the defined anchor. However, several broader claims about proofs, optimality and probabilities are false or overstated. The original files were not modified for this review.

The essential distinction is: **computational accuracy relative to a definition does not guarantee accuracy relative to nature.** A fixed anchor can be counted exactly even when the actual Sun deviates from it.

Area	Finding
Cumulative rounding	Proven: at most q/2 position error; less than q duration error between any two boundaries.
529 years / 23 residue classes	Proven for year-start residuals and chart-start residuals, respectively. Not for the complete cycle-residual state.
Vanishing mean residual	False: annual position residuals have an exact mean of 3/529 q.
Sturmian sequence	False in the usual mathematical sense; equal-length annual blocks provide a counterexample.
Global minimax optimality	Not guaranteed; on a test case, the implemented restricted search returns a provably inferior value.
Polar-field probability	98.7% is a calculation for a cell-centred forecast. Observed: 303/407 one-day field hits = 74.45%.
4 × 23 or 6 × 23	Both can be independent groupings. For a quarter-like level, 4 is the simpler choice.

Basis of the review

The review covered the 19-page PDF Integer Time Counting on a 1/23 Scale and the 2,738 lines of zyklus23(1).py. It examined proofs, exact counterexamples, existing verification functions and the complete circle study. Astronomical results were reproduced using the ephemerides employed by the project; this does not establish an independent comparison with nature. The Swiss Ephemeris comparison was not repeated in this environment. Details appear on page 6.

1 · What is mathematically proven

Unit and structure

```
q = 86400 / (768 · 66 · 23) s = 75/1012 s
q/2 = 75/2024 s ≈ 0.0370553 s
```

768 × 66 ticks correspond to exactly 86,400 seconds. A calendar cycle additionally contains its zero crossing and therefore lasts 1,519 or 1,520 q. Consequently, the system day is longer than an SI day. The PDF already states this difference correctly: it is a deliberate definition, not a rounding error.

Rounding and interval bounds

```
R(x) = floor(x + 1/2), ρ(x) = x − R(x)
−1/2 ≤ ρ(x) < 1/2
B(c) = R(τ(c/Z)), Z = 365 · 768 = 280320
```

Each cycle boundary is therefore at most q/2 from its target. For two boundaries a and b, the following identity holds exactly:

```
[B(b) − B(a)] − [τ(b/Z) − τ(a/Z)] = ρ(a) − ρ(b)
|duration error| < 1 q
```

This bound is independent of the length of the interval. It concerns rounding of the target trajectory; it does not bound astronomical model errors.

529-year period and chart residue classes

```
τ(x) = 425806080 x + (703993 x − 11 x²)/529
τ(n+529) − τ(n) = 529Y + b + g(2n+529) ∈ ℤ
```

For integer n, year-start residuals are exactly 529-periodic. A period p requires 529 | 2gp. Since gcd(22,529) = 1, this implies 529 | p; thus 529 is the smallest positive period. At n = 529k, the residual is zero.

At chart starts n = 23j, the fractional contribution is b·j/23. Since b modulo 23 equals 9, the 23 residue classes from −11/23 to +11/23 are each visited exactly once. **These chart-start residuals have an exact mean of zero.**

Calendar validity

Over the intended 966-year horizon, exact target increments per cycle lie between 1,519.0046041 and 1,519.0047475 q. Rounding can therefore yield only 1,519 or 1,520 q. Telescoping the annual sums gives the number of extra quanta. The parabola is not an indefinitely valid calendar: its slope continues to decrease. Any later extension requires an explicitly defined continuation.

2 · Three manuscript claims to correct

A. The mean position residual does not vanish

PDF pp. 5-6, Definition 4 and Theorem 3(iii), and Python documentation line 44. Exact rational arithmetic over one complete annual-residual period gives:

```
Σ[n=0..528] ρ(n) = 3 q
Mean = (3/529) q ≈ 0.420287 ms
```

Because of the 529-year periodicity, this mean persists over arbitrarily many complete periods. The PDF's reported mean of +0.0009 q over 967 year starts is reproducible, but does not prove a vanishing long-term mean.

Appropriate correction: the average duration error per unit vanishes because the total interval error remains bounded:

```
{[B(N)−B(0)] − [τ(N/Z)−τ(0)]}/N
= [ρ(0)−ρ(N)]/N → 0
```

The substantive claim of drift-free rounding is therefore retained. "Quantum irrelevance" may be used as a defined term, but should require vanishing mean duration error rather than vanishing mean position residual. It describes a property of the counting quantum q.

B. The extra quanta do not form a Sturmian sequence

PDF p. 5. Sturmian sequences are balanced: equal-length subblocks differ in their number of ones by at most one. Here, two complete annual blocks each contain 280,320 cycles, but have 1,331 and 1,290 extra quanta, respectively. The difference of 41 disproves this property.

Appropriate correction: "Cumulative rounding distributes the extra quanta approximately uniformly on a local scale." Two adjacent spacings within individual years do not establish a global Sturmian property. A constant rational slope produces a periodic mechanical sequence; a constant irrational slope can produce a Sturmian sequence.

C. The full residual state does not repeat after 529 years

The abstract makes a broader claim than the valid Theorem 1. For cycle c = 1, calculation gives:

```
ρ(1 + 529Z) − ρ(1) = −22/Z = −11/140160 q ≠ 0
```

Appropriate correction: "The year-start residuals repeat exactly after 529 years." Year lengths themselves do not repeat either: the annual surplus J changes by −22 q after 529 years. A residual period is not a complete calendar period.

3 · Implementation errors and optimality

A. The minimax search has no global guarantee

Python lines 365-391. For fixed g, the objective is convex in b. However, this does not justify searching g only near its least-squares value. An exact counterexample is:

```
e = [0, 1, −1, 1], denominator = 529
Returned: b = −132, g = 79
Maximum error = 582/529 ≈ 1.100189
Better choice: b = 0, g = 0; maximum error = 1
```

An error below 1 would require p(n) = n(b+gn)/529 to satisfy p(1) > 0, p(2) < 0 and p(3) > 0. The linear function b+gn cannot have that sign pattern. Hence error 1 is globally optimal, and the search misses it.

The actual solar coefficients b = 703,993 and g = −11 are reproduced. This counterexample does not show that those particular coefficients are suboptimal; it disproves the general guarantee. A complete proof could use integer optimisation with certified bounds, or a provably exhaustive search domain.

B. Specific boundary errors

Location	Finding and required correction
jd_zu_text(), lines 209-224	Just before midnight, the function can produce “2000-01-01 24:00:00”. A carry from rounding seconds must also update the date.
SplineAnker, lines 285-305	Horizons of 1, 22, 24 or 25 years with step size 23 can fail with TypeError. Handle the remaining segment or validate allowed horizons in advance.
jd_tt_aus_utc(), lines 813-823	TT−UTC is assumed to be 69.184 s for every year. In 2000 it was 64.184 s: an error of +5 s. Use a historical leap-second table or restrict inputs.
gezeiten(), lines 1601-1613	S2 is calculated from UT1, so its SI period is not exactly constant. At t=309814735349 q, S2(t)=763788, but after 582912 q it is 763787. Restrict “exact” to the ideal grid.
All coordinates zero at jj	True for the calendar. Lunar phase is 65331, and the rotation axis also starts at a nonzero value. Specify which axis is meant.

The additional four-part or six-part grouping requires no changes to these computational cores. It can be derived from the integer chart index.

4 · Statistics and astronomical interpretation

Polar fields: cell centres versus actual hits

PDF pp. 10-11 and bericht_pol(), particularly line 2303. For its P(field) calculation, the table places the forecast centre at (0,0), always at a cell centre. Actual forecast centres can lie near cell boundaries. The probability in the nearest field must be integrated separately for each actual centre.

Horizon	PDF / centred	Mean model P	Observed field hits
1 day	98.67%	74.05%	303/407 = 74.45%
30 days	1.20%	1.19%	11/406 = 2.71%
365 days	0.167%	0.166%	1/383 = 0.261%

Depending on position relative to cell boundaries, the correct one-day model probability ranges from 29.38% to 98.64%. This does not require changing the forecast mean; the evaluation and labels need correction. The hindcasts also yield RMSE rather than bias-corrected standard deviations. Gaussian shape, independence of the axes and forecast calibration are additional assumptions.

At the one-year horizon, the nominal 68% radius actually contains 238/383 = 62.14% of cases; the 95% radius contains 370/383 = 96.61%. These are descriptive results from the same data; overlapping horizons generate dependent errors. Independent calibration assessment on a separate time period remains necessary.

Definition, ephemeris model and observation

The solar derivation reproduces a maximum deviation of 57.5 s from the smoothed reference and residual scatter of 0.71 min after eight terms. For the Moon, 3.2 s relative to the smoothed phase model and 3.2 min residual scatter after eight terms are reproduced. These are different comparison quantities. Even an exactly counted anchor may miss the actual Sun by minutes or an actual new moon by hours.

The planned final calibration using DE440 cannot cover the entire horizon: JPL specifies 1550-2650 for DE440, whereas the project extends to 2995. DE441 covers this interval. The ephemeris and time scale must be selected before final calibration [2].

The statement about UTC from 20 May 2027 is consistent with CGPM Resolution 5 adopted on 15 October 2026 [1]. It does not remedy the historical UTC-conversion error.

Circle study

Reproduced: 24 circles for the Sun at 1″ RMS, 109 for the Moon at 1″, and 113 for Mars at 1′; 140 circles beat the fitted Mars Kepler error. This applies to the particular search method and fitting windows. It proves neither the smallest possible number of circles nor predictive ability. The Kepler and circle fits also optimise different error measures: Cartesian positions and angles, respectively.

5 · Intermediate cycles, priorities and evidence

4 × 23 or 6 × 23

Interpretation: four or six charts of 23 years each. These are additional phase groupings; their mathematical validity does not prove astronomical usefulness.

Option	Length	Structural use	Remainder in 529 years
4 × 23	92 years	Four phases; two opposite pairs. Simplest quarter-like level.	5 full groups + 3 charts
6 × 23	138 years	Six phases; 3 pairs or 2 groups of three.	3 full groups + 5 charts

```
j = floor(n/23) # chart index
group, phase = divmod(j, m), m = 4 or 6
```

Recommendation: use m = 4 as an additional display level that runs independently. Retain the 529-year anchor. The grouping phase and annual-residual phase recur jointly after 2,116 years for m=4 and 3,174 years for m=6. These figures describe label and residual phases only, not repetition of actual year lengths. Restarting the grouping at every great-chart boundary would create incomplete final groups.

Suggested order of work

1. Correct Theorem 3, the Sturmian label and the abstract. **2.** Replace polar-field evaluation with calculations using the actual forecast centres. **3.** Establish the minimax guarantee or narrow the claim. **4.** Fix date rollover, historical UTC conversion and remaining spline segments. **5.** Specify the astronomical reference and its validity interval. **6.** Add the optional four-part grouping and assess its value for a concrete application.

Verification performed and limitations

The unchanged main program completed all available verification blocks successfully. These included 39,825 standard-calendar round trips, 20,009 lunar round trips and four older calendar modes with approximately 23,065 cases each. Exact counterexamples, the full 150-circle study and polar-field evaluation were also examined. These samples supplement the proofs; they do not replace an exhaustive proof.

Environment: Python 3.12; NumPy 2.3.5; PyEphem 4.2.1; astropy-iers-data 0.2026.11.2.0.34.55; ERFA 2.0.1.5; SciPy. IERS data: 23,648 days, MJD 37665-61312. Swiss Ephemeris could not be installed; the claimed house-cusp deviation of 3.1″ was not independently checked here. No new DE440/DE441 fit or independent astronomical observation dataset was used. No claim of methodological novelty was established.

Primary sources and references

[1] BIPM, CGPM 2026, Resolution 5: bipm.org/en/cgpm-2026/resolution-5
[2] Park et al. (2021), JPL, DE440/DE441: ssd.jpl.nasa.gov/doc/de440_de441.html
[3] IERS Bulletin C 20: TAI−UTC = 32 s from 1999: datacenter.iers.org/data/html/bulletinc-020.html
[4] IERS, TT from TAI: TT(TAI) = TAI + 32.184 s; PyEphem Quick Reference: rhodesmill.org/pyephem/quick.html